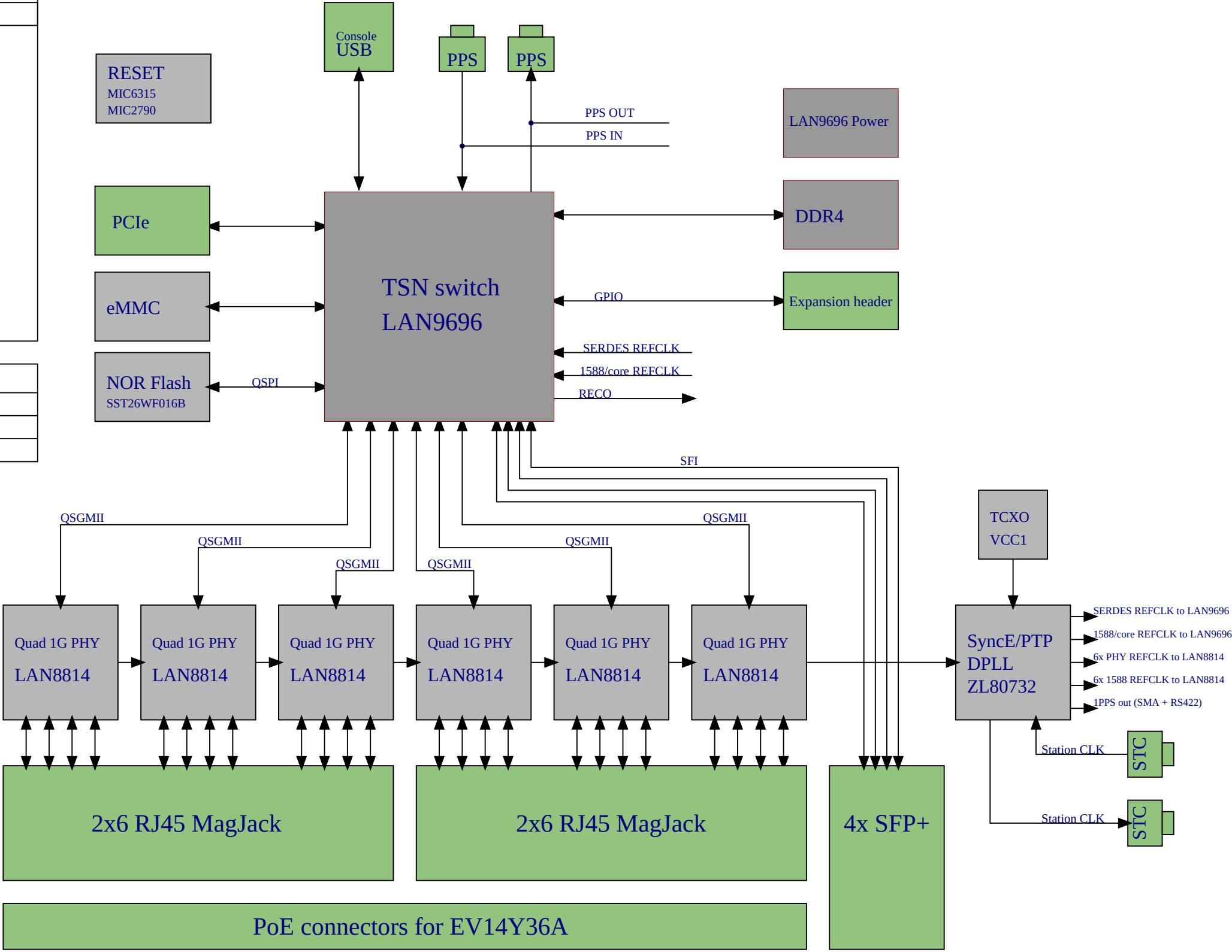
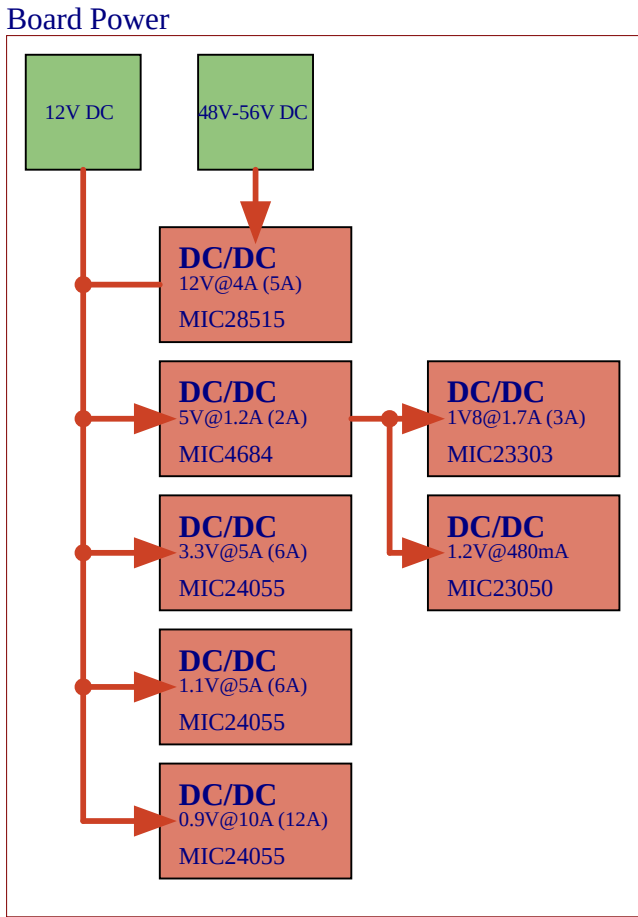


Block Diagram & TOC

Table of Contents	
Sheet	Description
1	Block Diagram & TOC
2	LAN9696 System Control, Memory
3	LAN9696 Ports
4	Copper Ports &POE
5	LAN8814 Quad PHY
6	POE Magjack
7	SFP
8	ZL80732 DPLL and Timing
9	PCIe, RGMII and USB
10	DDR4
11	LAN9696 Power
12	Board Power
13	Regulator Modules

Default Selections	
1	Resistors are 1%, 100mW, 0402
2	Capacitors are X5R, X7R, or C0G, 50V, 0402



Revision History			
Revision	Date	Revision Summary	Author
C0	2024/03/25	Seed build release	Jorgen Abrahamsen
R1	2024/05/16	Production release with BOM updates and ECO information	Jorgen Abrahamsen

Project Owner:
Jorgen Abrahamsen

PCB Layout Contact:
Jorgen Abrahamsen

MICROCHIP

PartNumber:
EV23X71A

Project Title
LAN969x 24-port EVB

Variant: Standard

Sheet Title
Block Diagram & TOC

Size A3

SCH #: 02-00733

Rev: 2

Date: 5/17/2024

PCB #: 04-11815

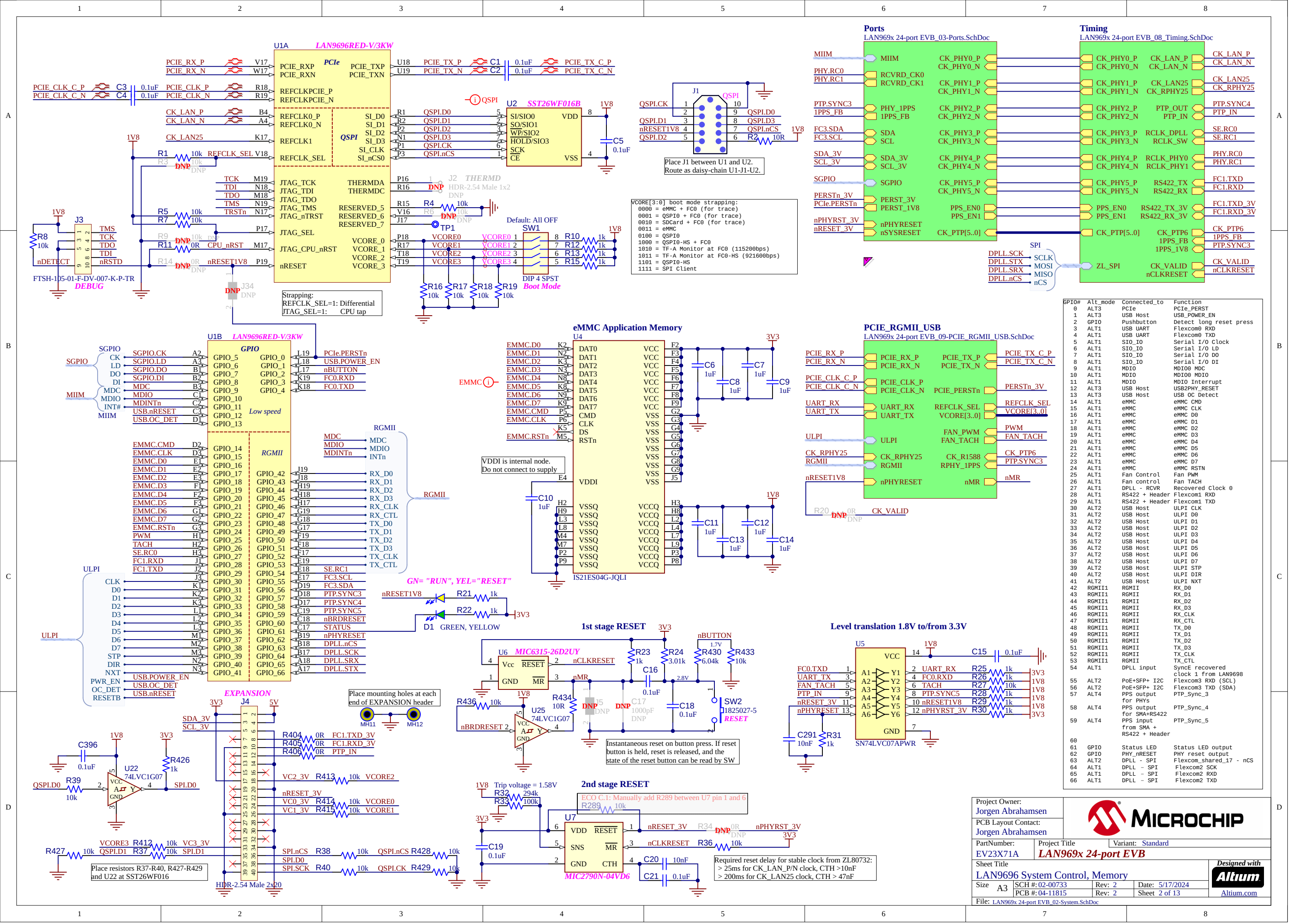
Rev: 2

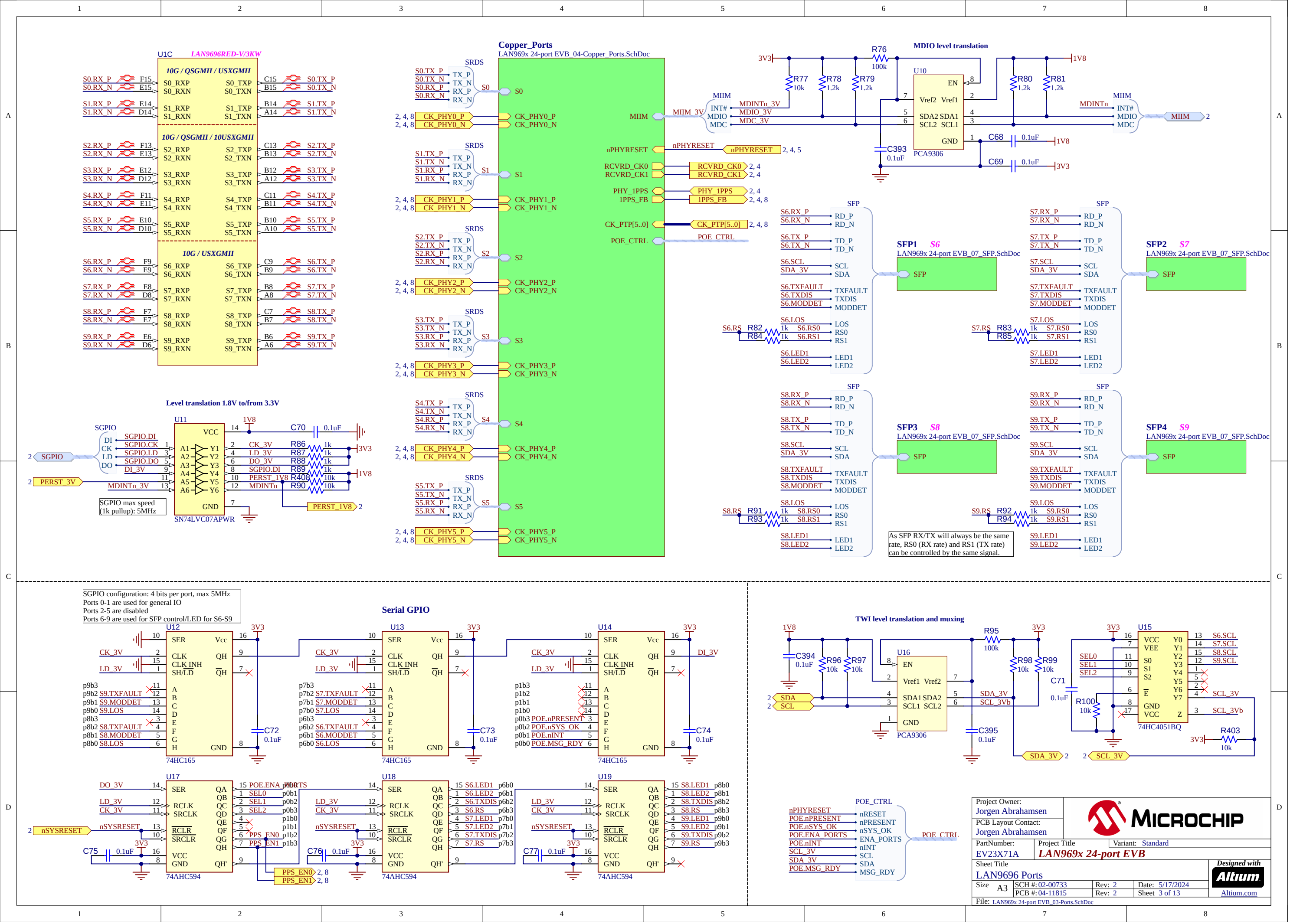
Sheet 1 of 13

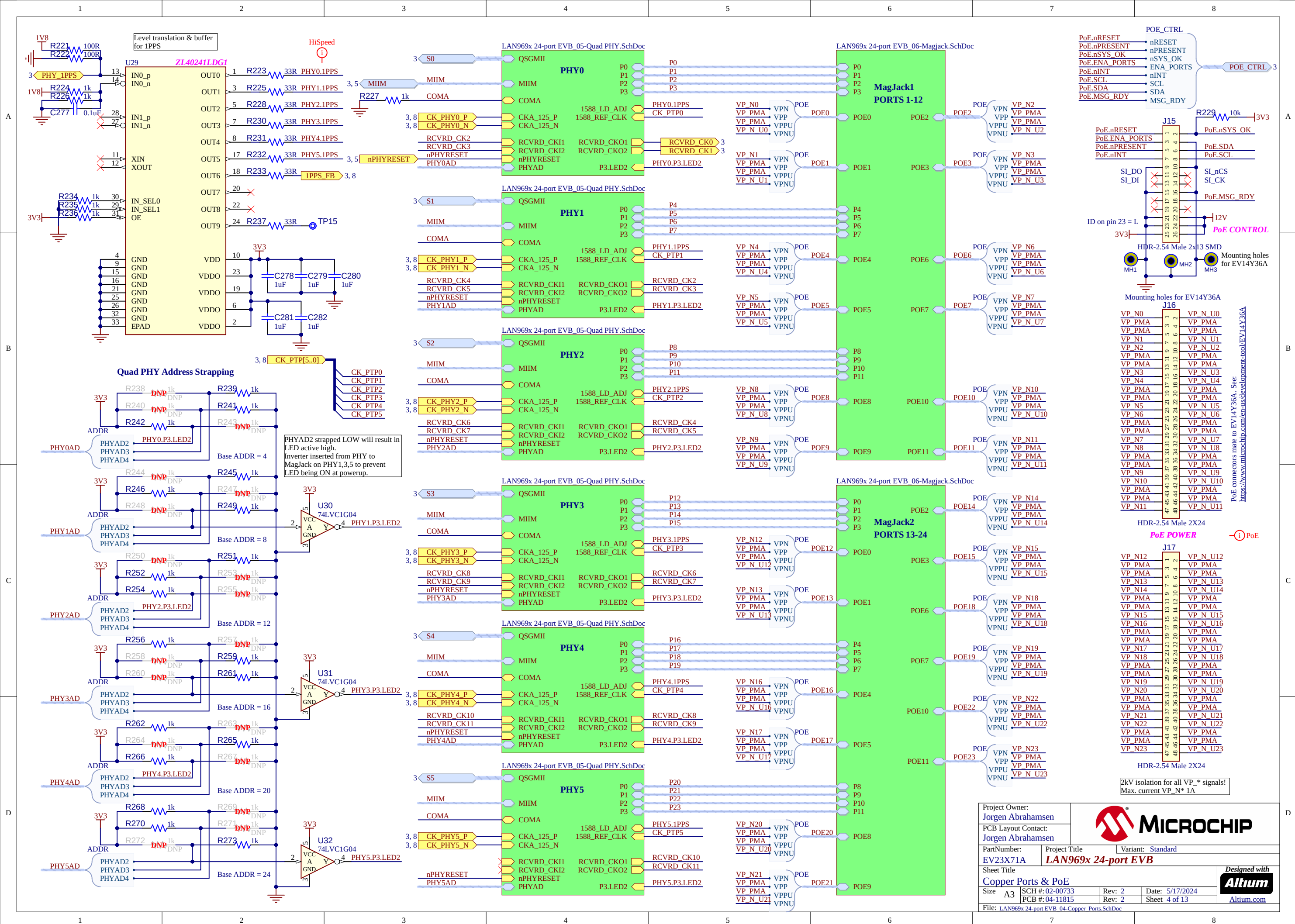
Designed with
Altium

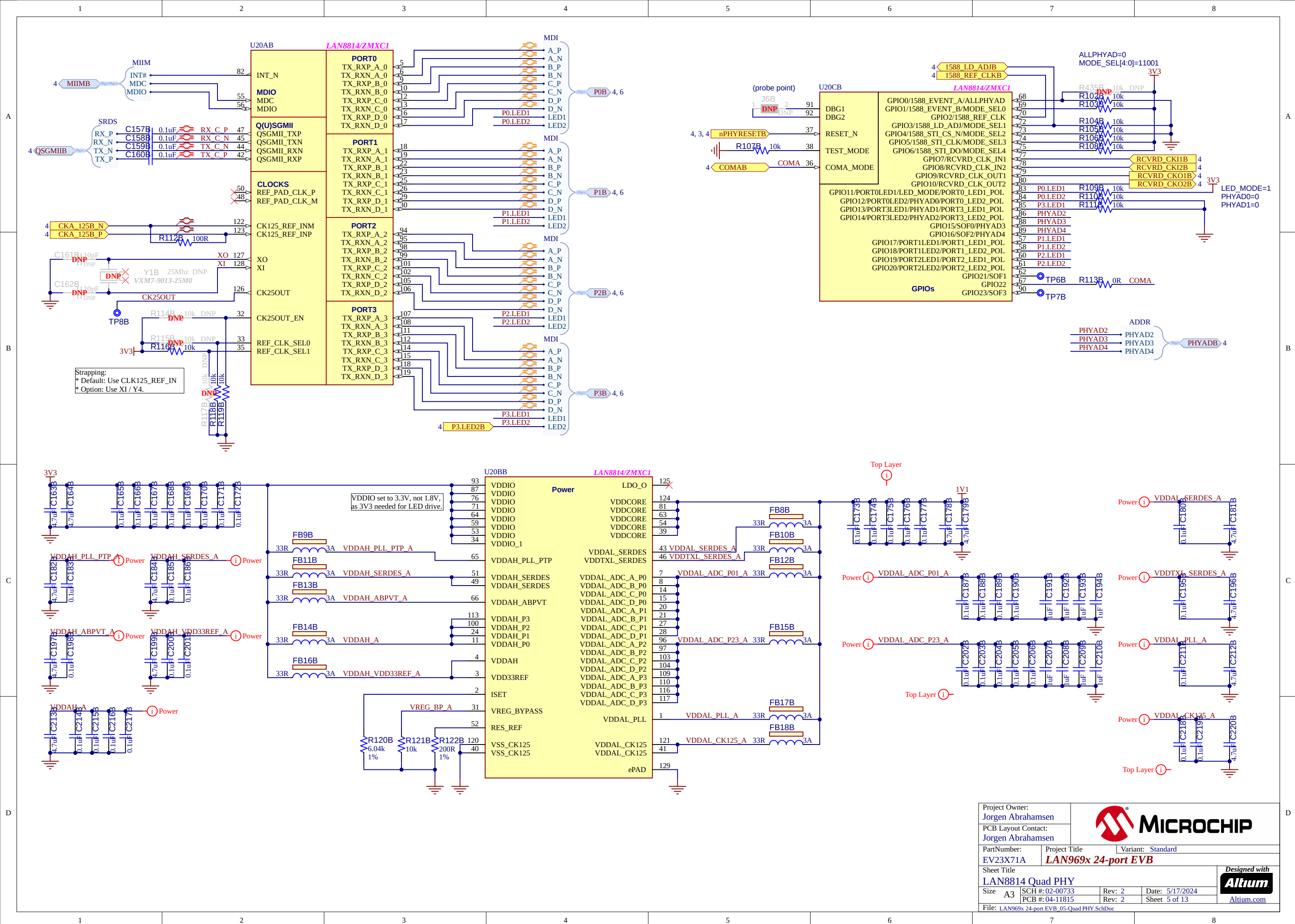
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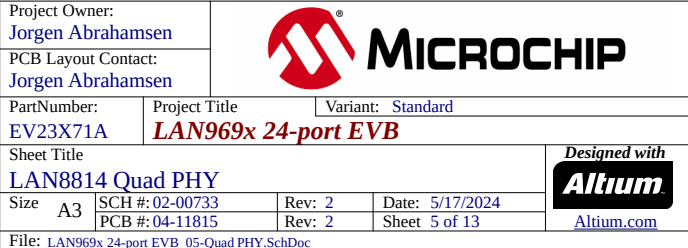
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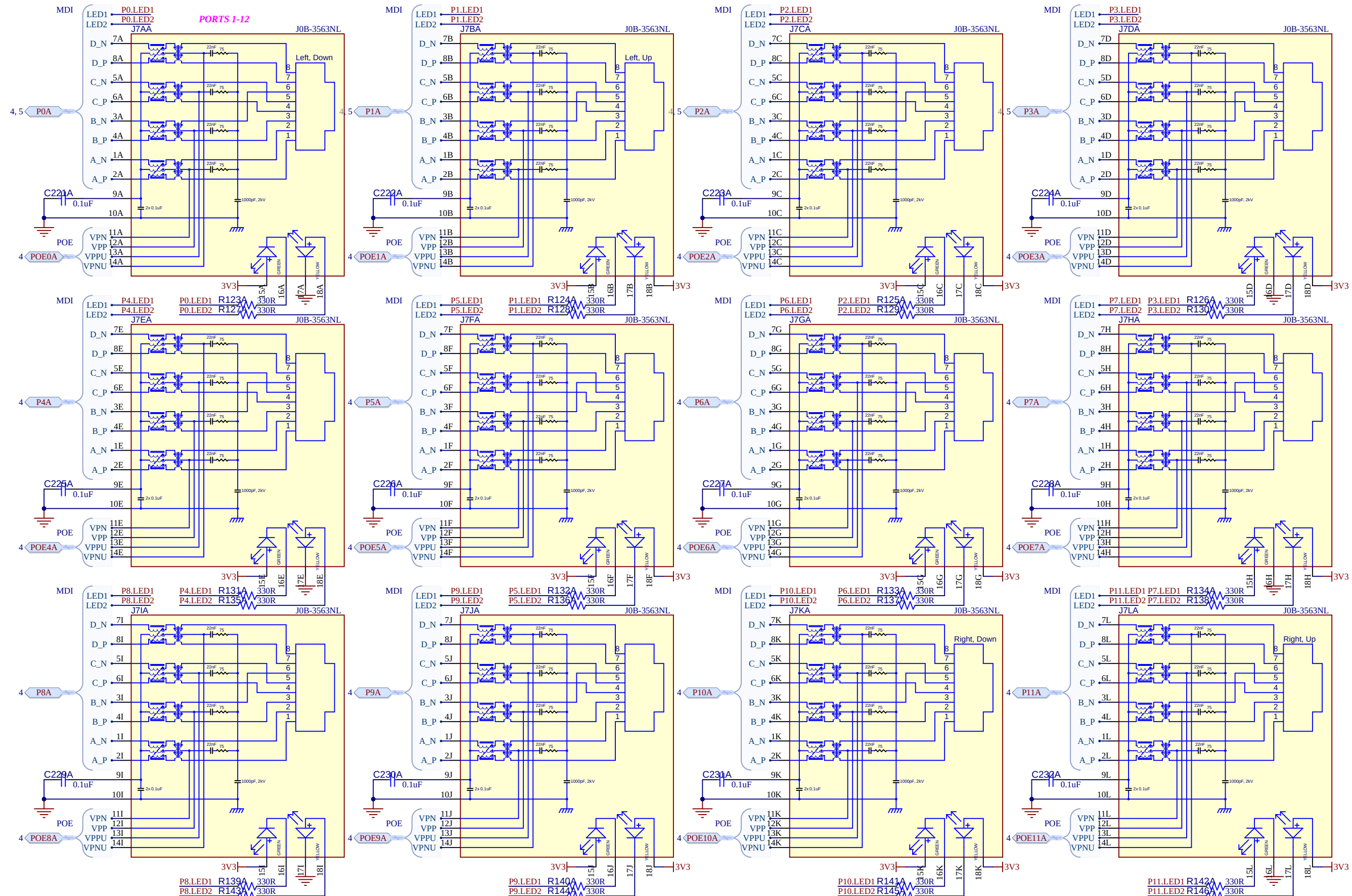






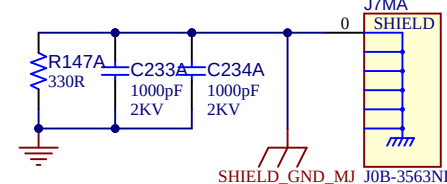




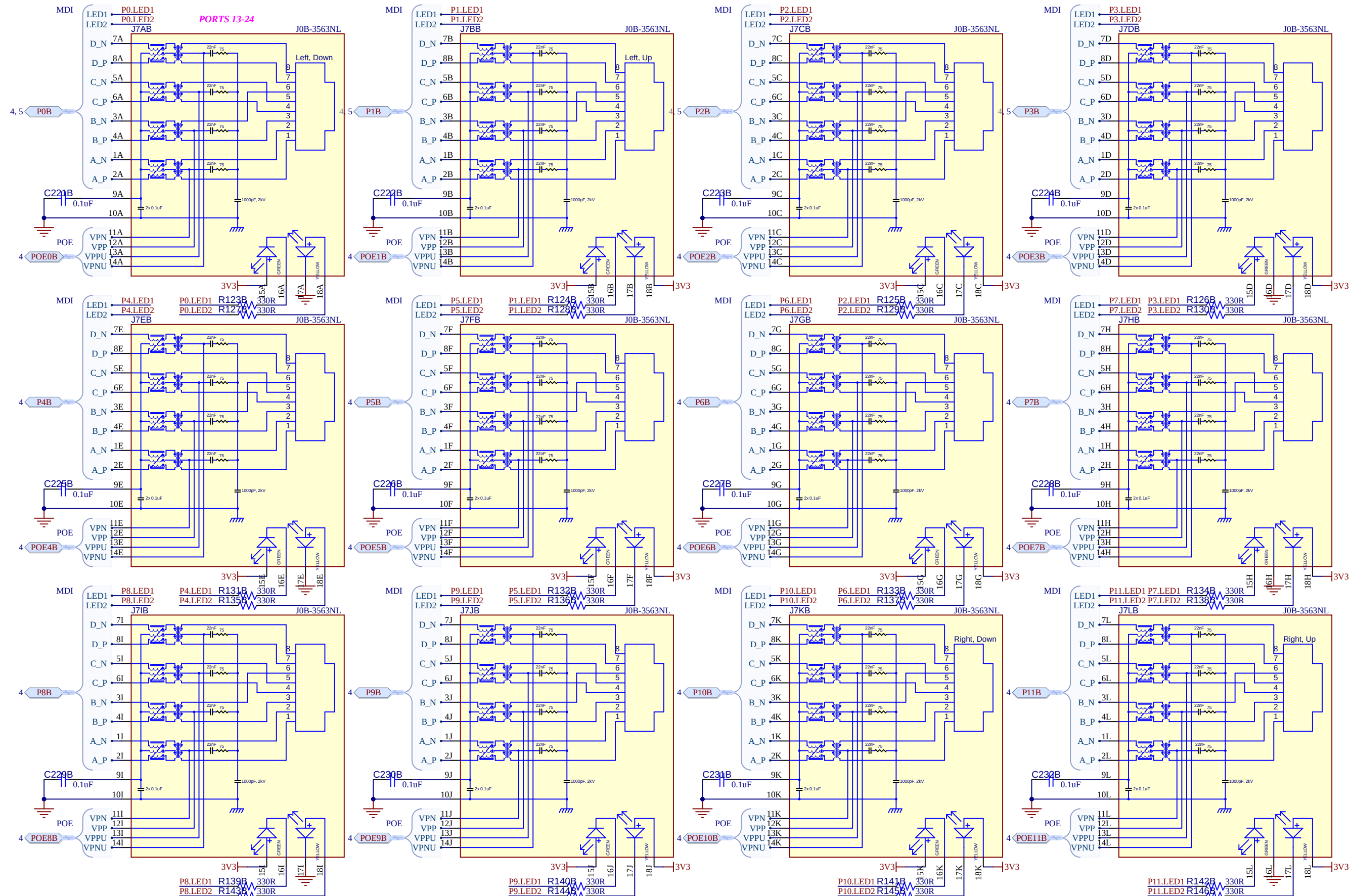


For each quad PHY, P0.LED2 and P3.LED1 will be active high, as PHYAD[1:0] are strapped low.

Power over Ethernet connected as per IEEE802.3.BT:
Alternative A (MDI-X): 1&2=Negative, 5&6=Positive
Alternative B(S): 4&5=Positive, 7&8=Negative

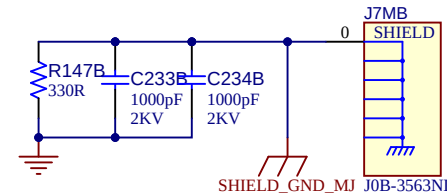


Project Owner: Jorgen Abrahamsen		 MICROCHIP	
PCB Layout Contact: Jorgen Abrahamsen			
PartNumber: EV23X71A	Project Title LAN969x 24-port EVB	Variant: <u>Standard</u>	
Sheet Title PoE Magjack		Designed with  Altium.com	
Size A3	SCH #: <u>02-00733</u> PCB #: <u>04-11815</u>		
File: LAN969x 24-port EVB_06-Magjack.SchDoc			

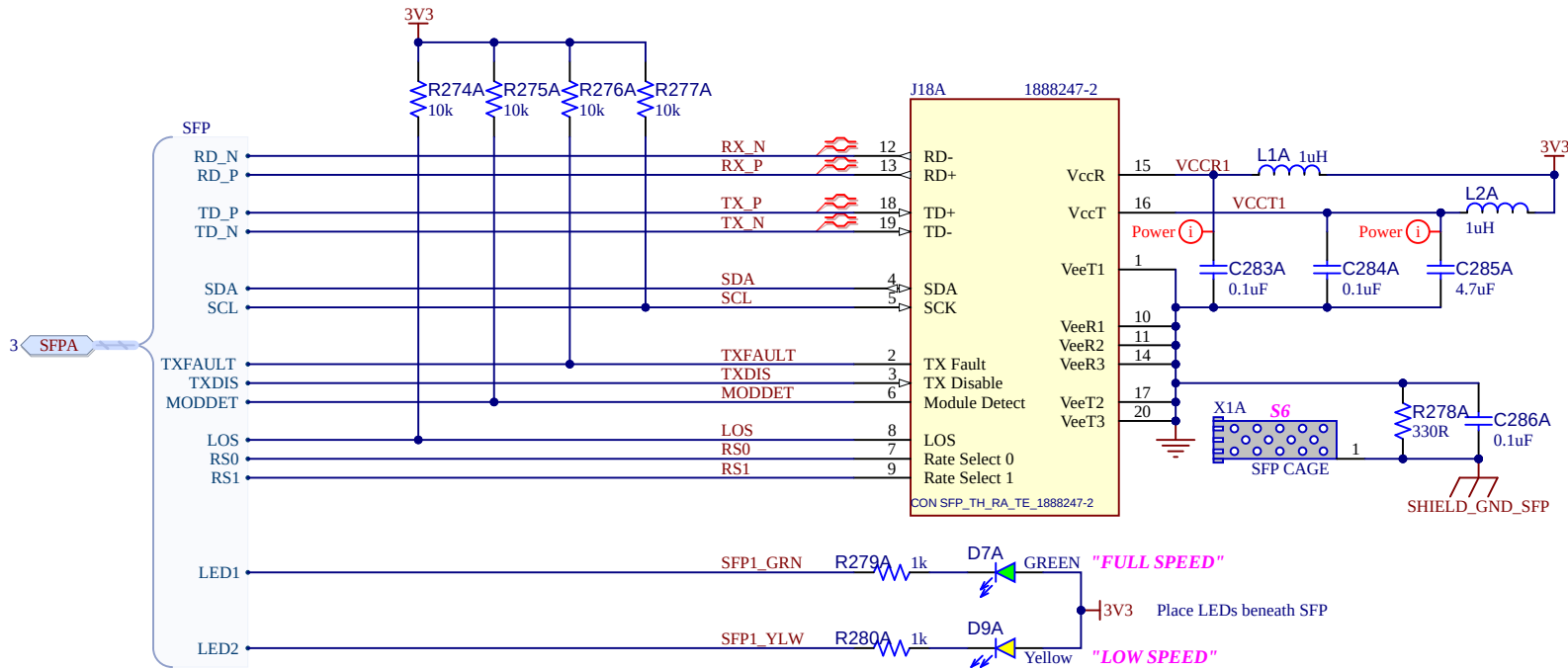


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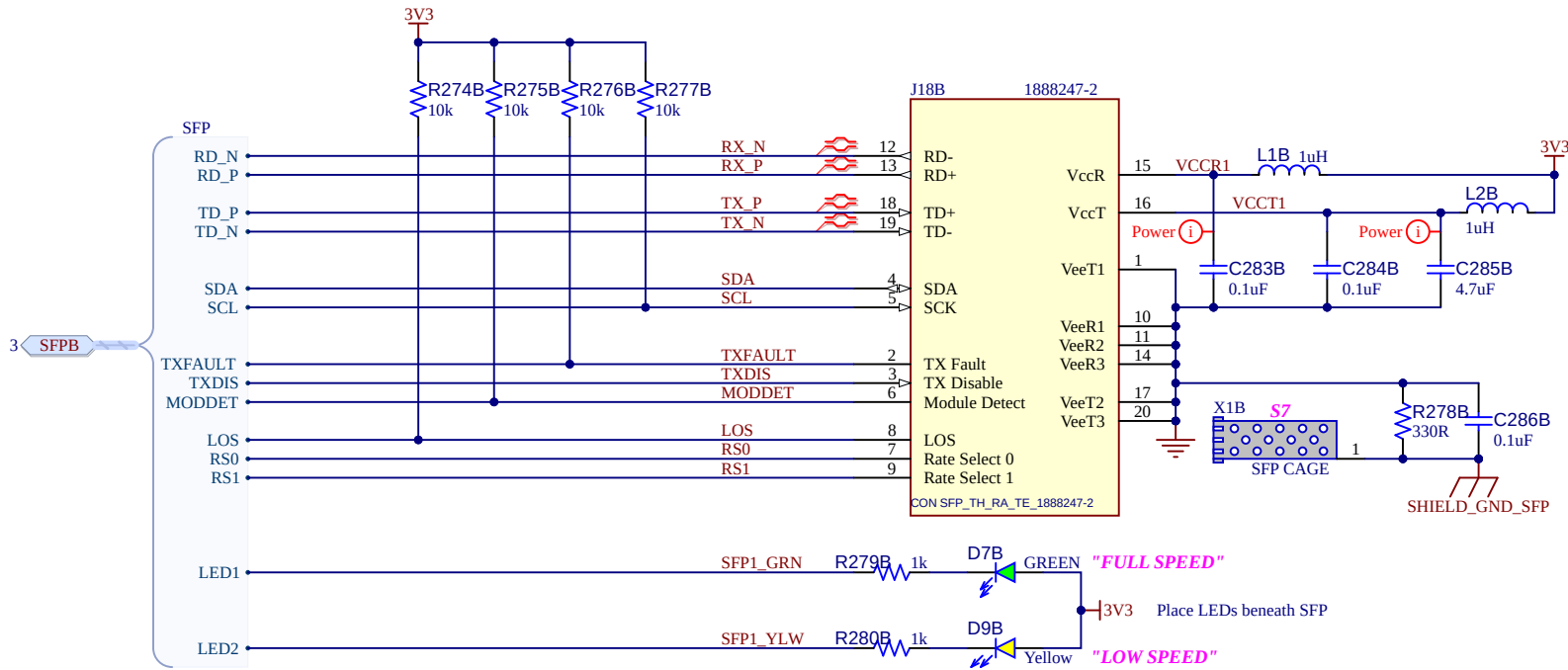
Power over Ethernet connected as per IEEE802.3.BT:
Alternative A (MDI-X): 1&2=Negative, 5&6=Positive
Alternative B(S): 4&5=Positive, 7&8=Negative



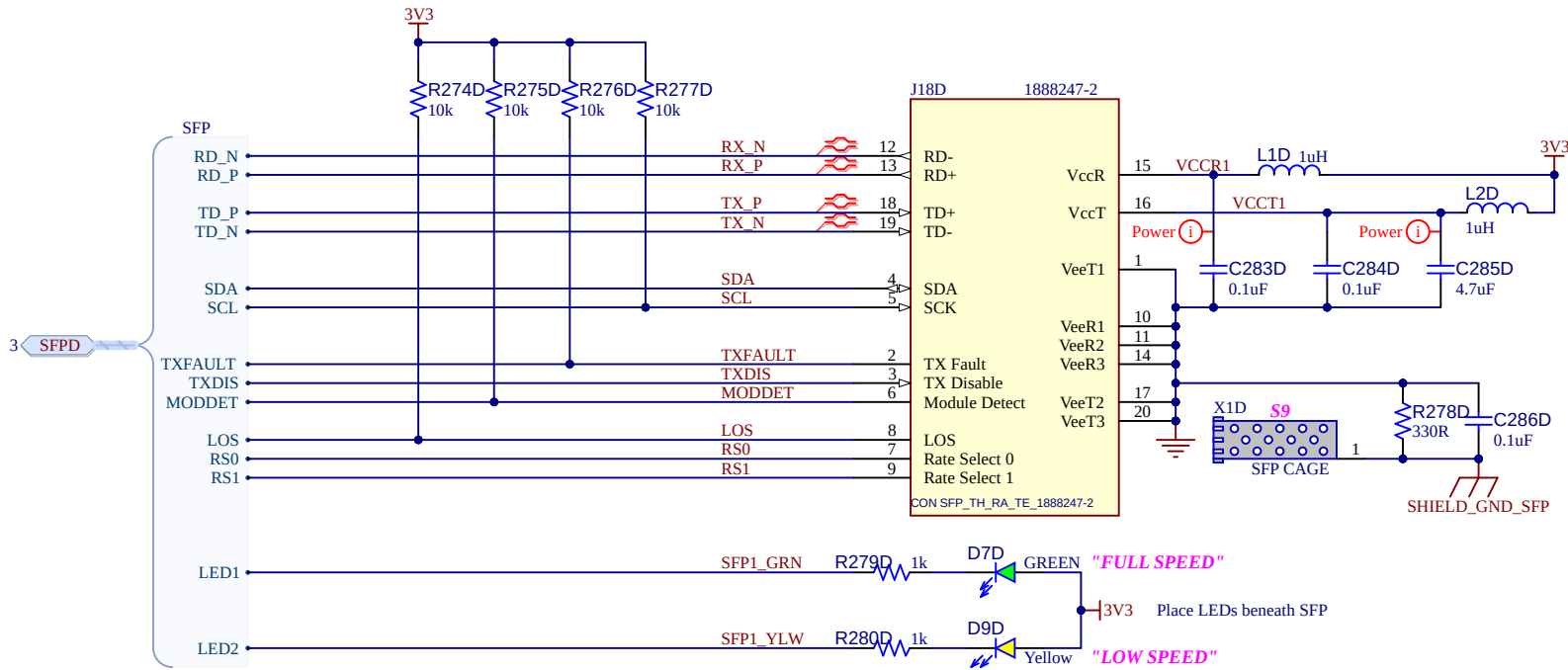
Project Owner: Jorgen Abrahamsen		 MICROCHIP	
PCB Layout Contact: Jorgen Abrahamsen			
PartNumber: EV23X71A	Project Title LAN969x 24-port EVB	Variant: <u>Standard</u>	
Sheet Title PoE Magjack		Designed with  Altium.com	
Size A3	SCH #: <u>02-00733</u> PCB #: <u>04-11815</u>		
File: LAN969x 24-port EVB_06-Magjack.SchDoc			



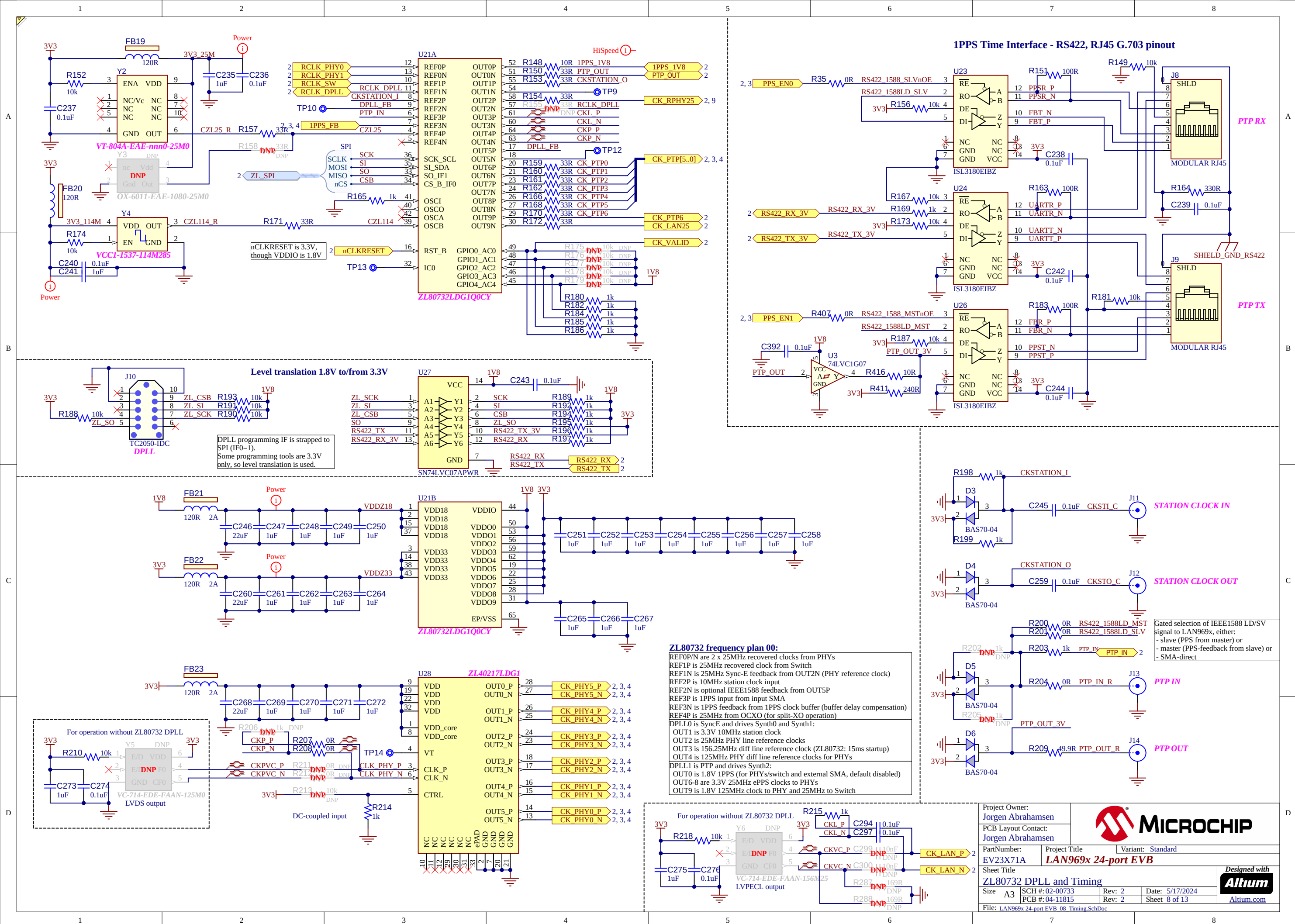
Parameter	State	Conditions
RS0	Low	RX signalling rate less than or equal to 4.25 GBd
	High	RX signalling rate greater than 4.25 GBd
RS1	Low	TX signalling rate less than or equal to 4.25 GBd
	High	TX signalling rate greater than 4.25 GBd



Parameter	State	Conditions
RS0	Low	RX signalling rate less than or equal to 4.25 GBd
	High	RX signalling rate greater than 4.25 GBd
RS1	Low	TX signalling rate less than or equal to 4.25 GBd
	High	TX signalling rate greater than 4.25 GBd

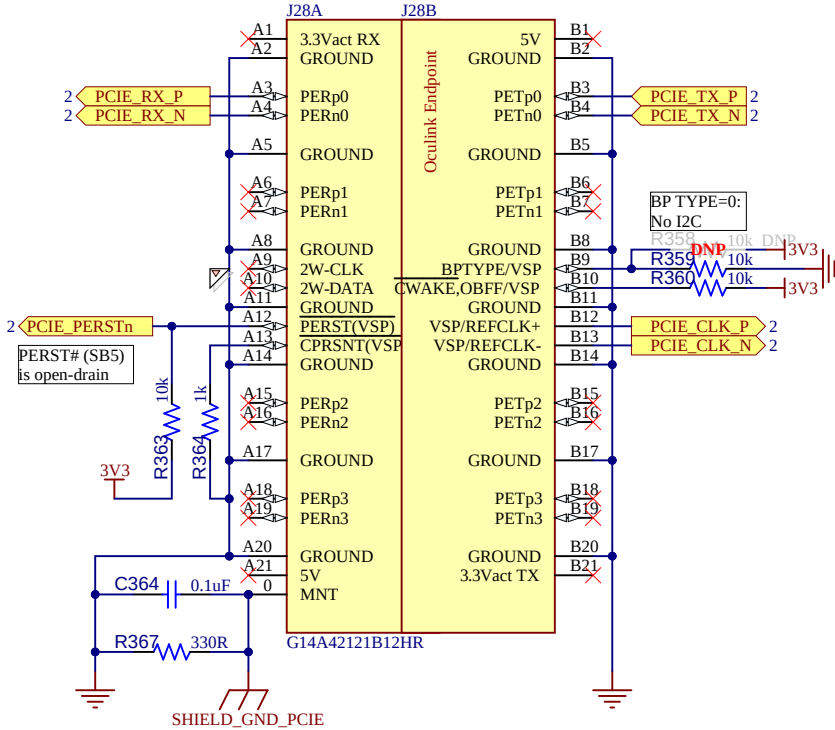


Parameter	State	Conditions
RS0	Low	RX signalling rate less than or equal to 4.25 GBd
	High	RX signalling rate greater than 4.25 GBd
RS1	Low	TX signalling rate less than or equal to 4.25 GBd
	High	TX signalling rate greater than 4.25 GBd

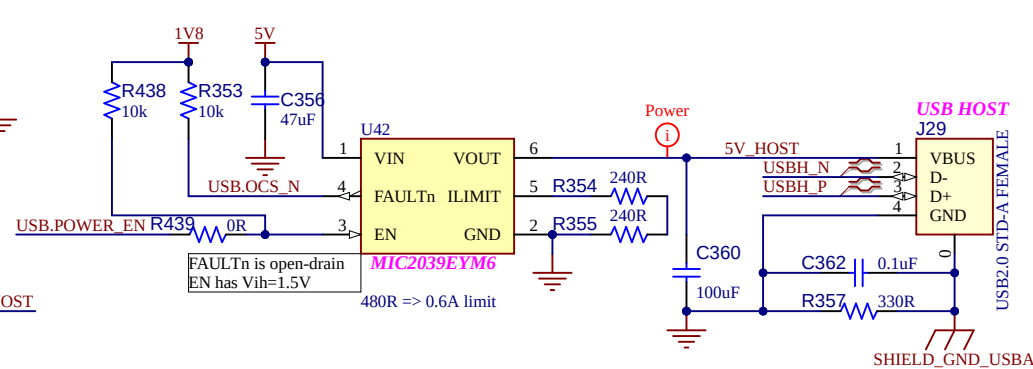
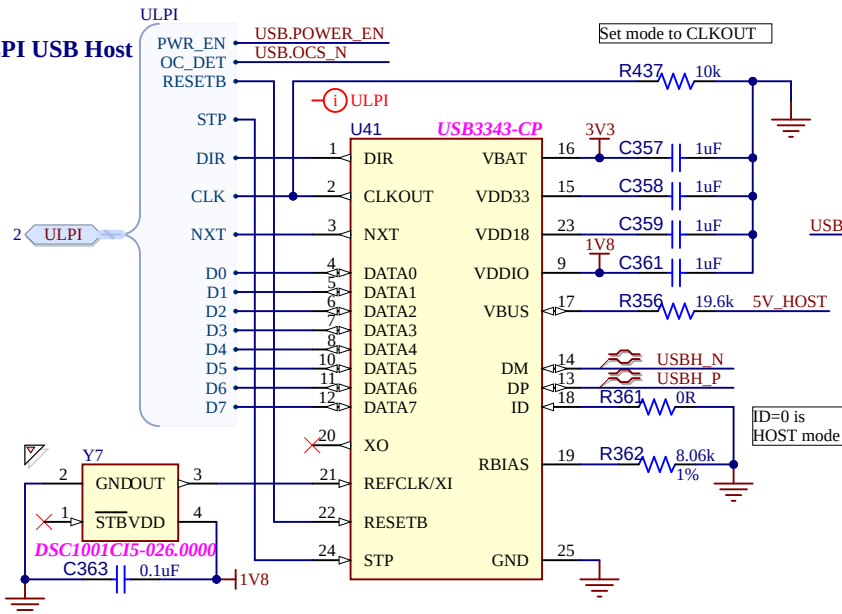


PCie - Oculink connector

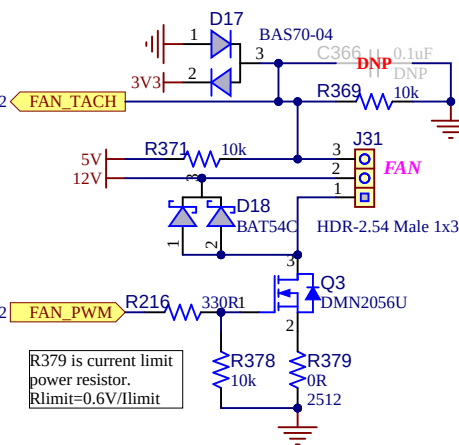
PCie



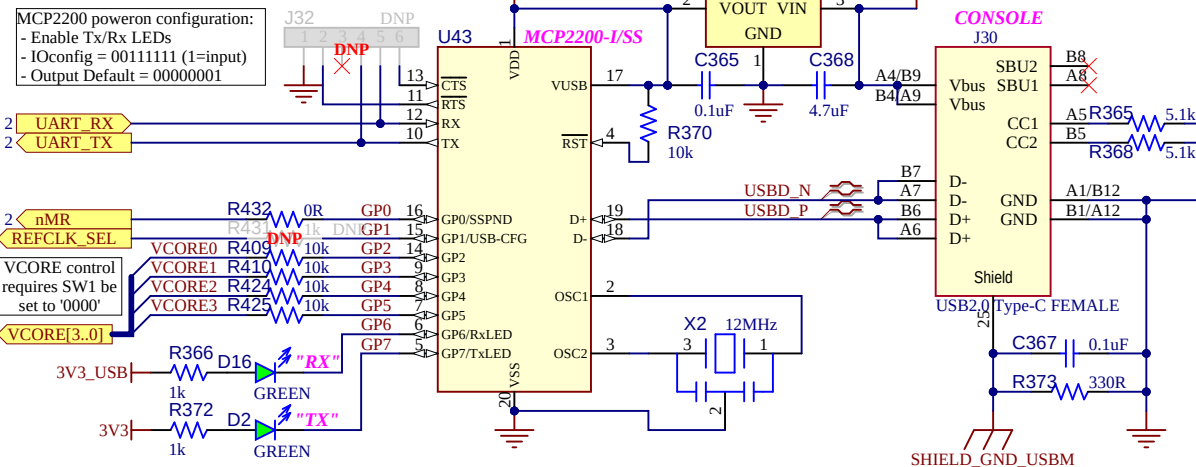
ULPI USB Host



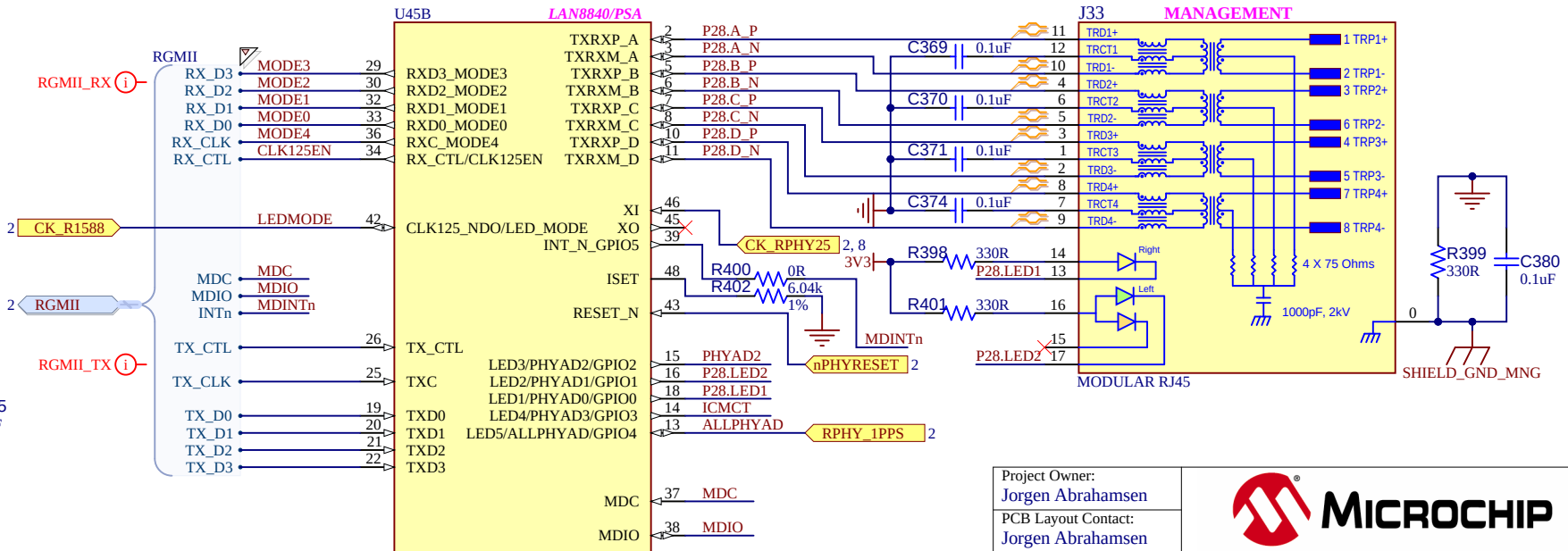
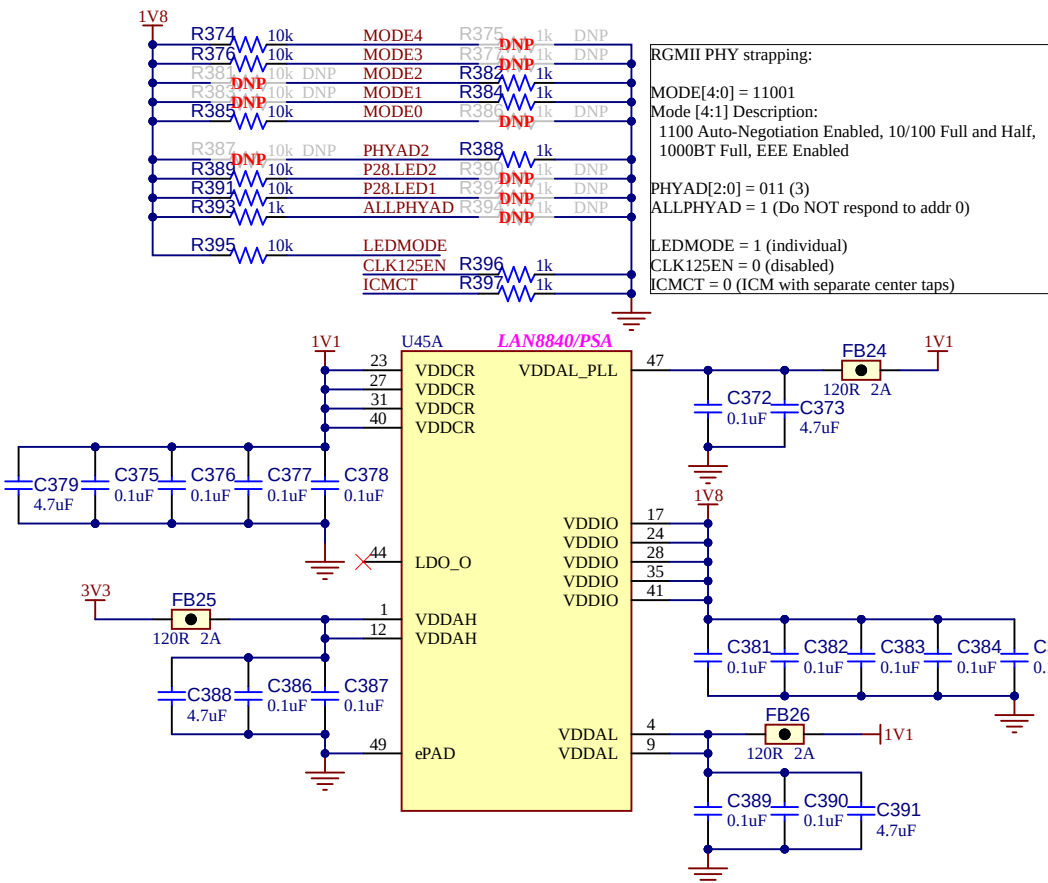
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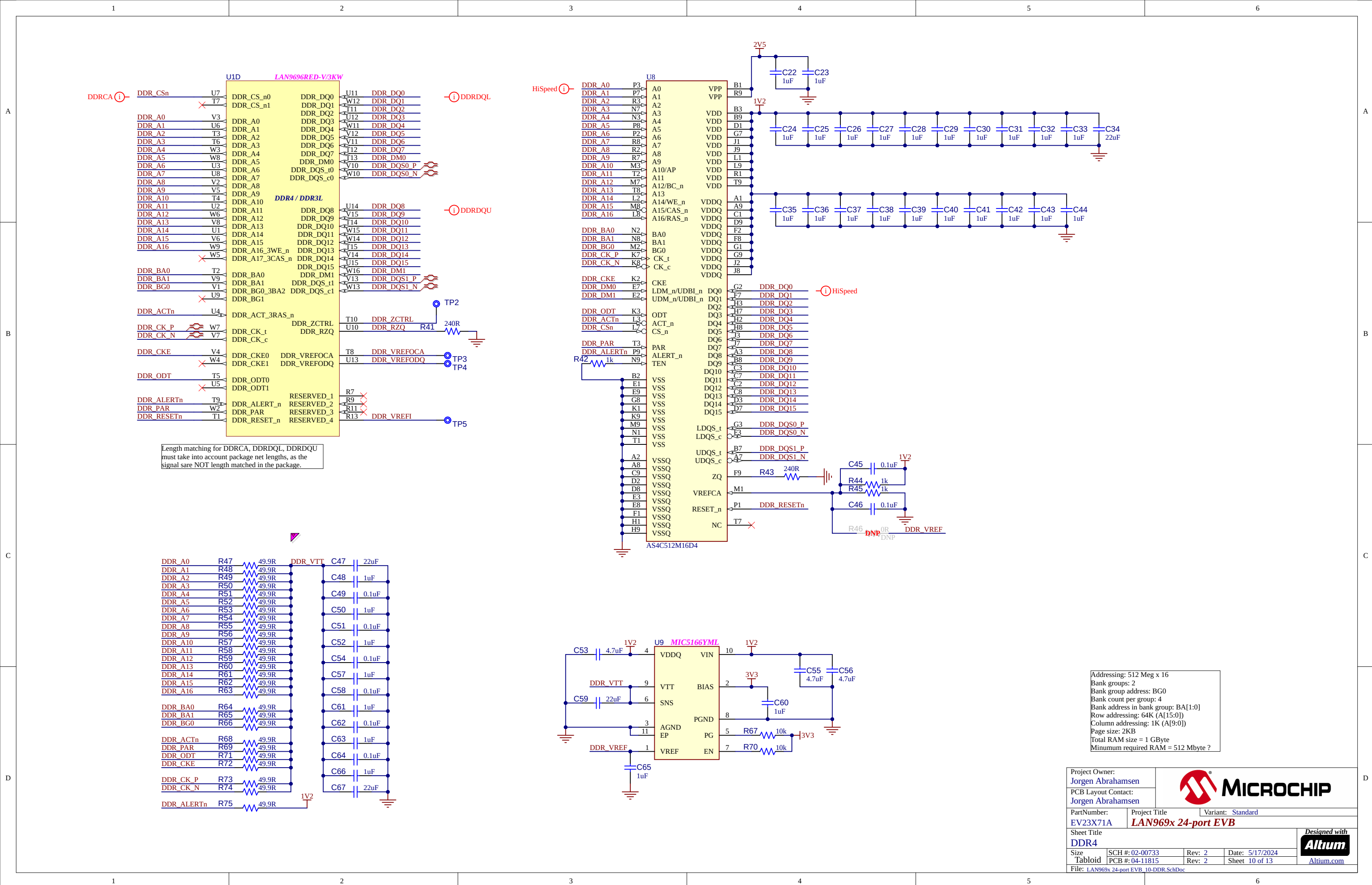


USB Console

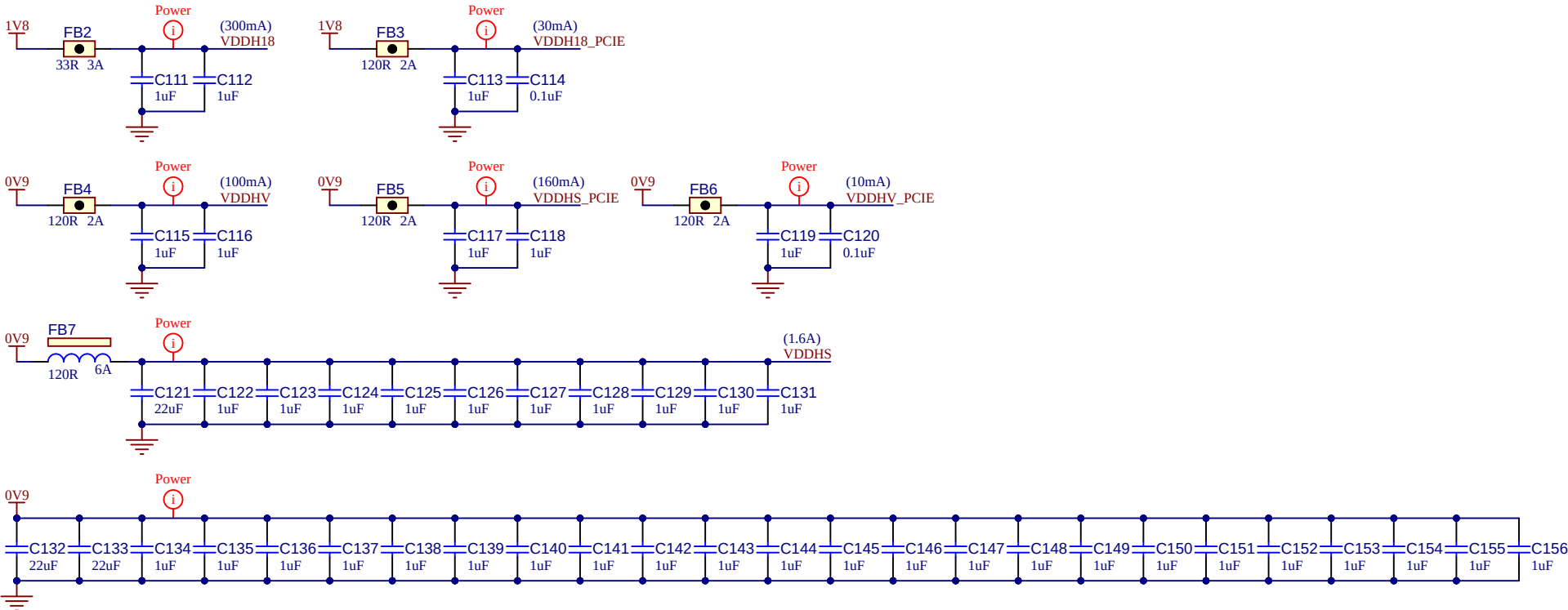
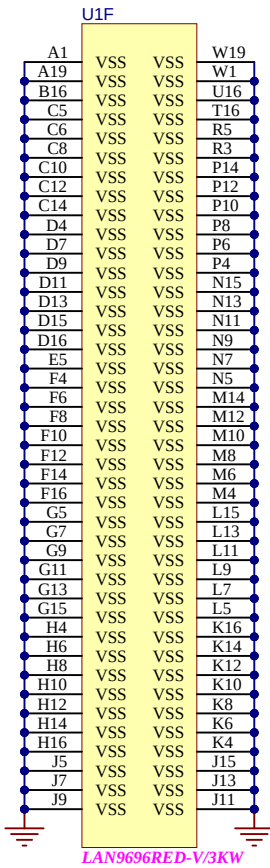
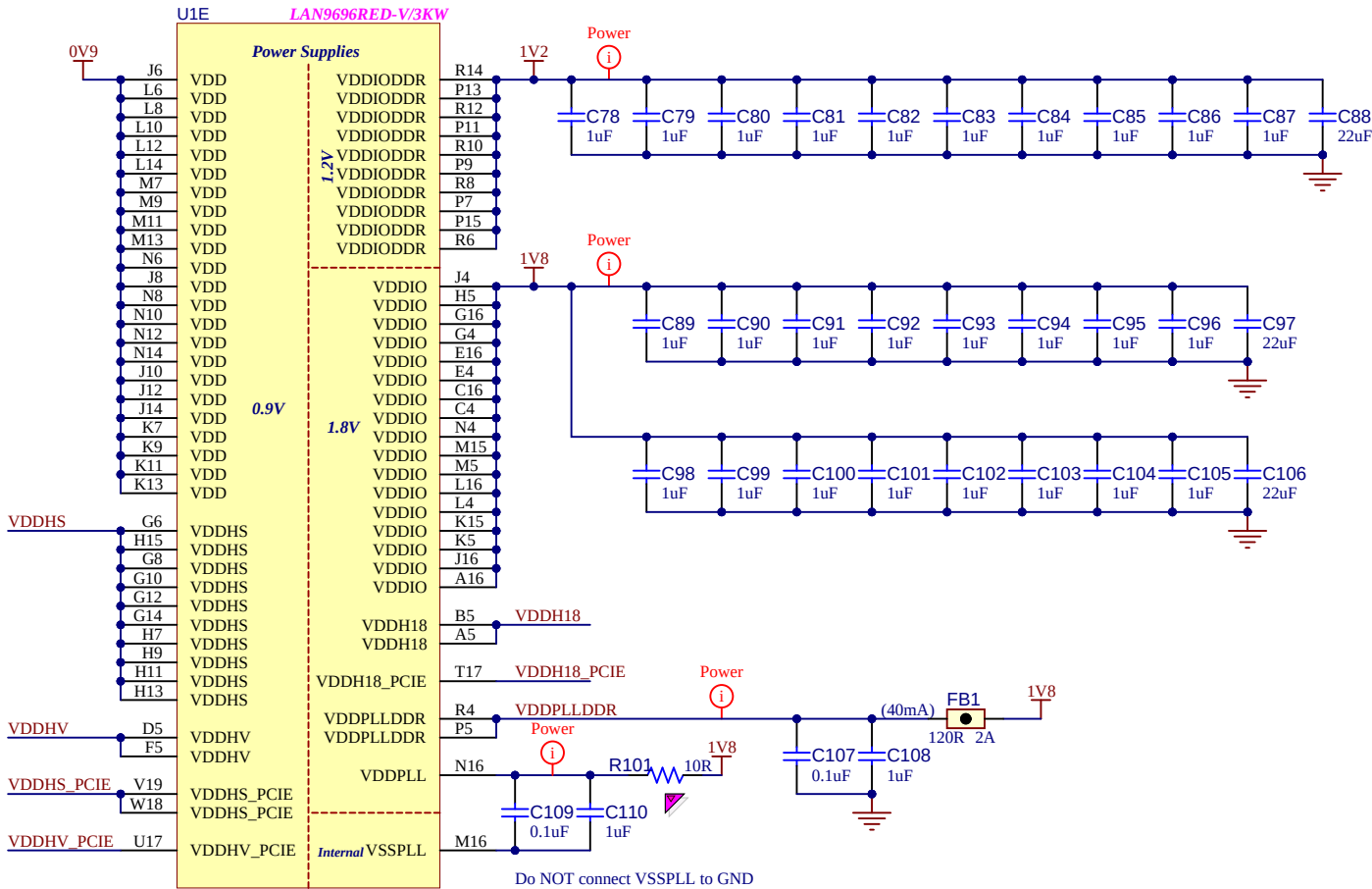


RGMII PHY: Management port





Power supplies:
0.9V 6.7A : LAN9698 VDD(5A)+VDDHS(1.6A)+VDDHV(0.1A)
1.2V 0.5A : LAN9698 VDDIODDR(0.2A)+DDR VDDQ(0.3A)
1.8V 1.7A : LAN9698 VDDH(0.3A)+VDDIO(0.5A)+ZL30732(0.5A)+Other(0.4A)

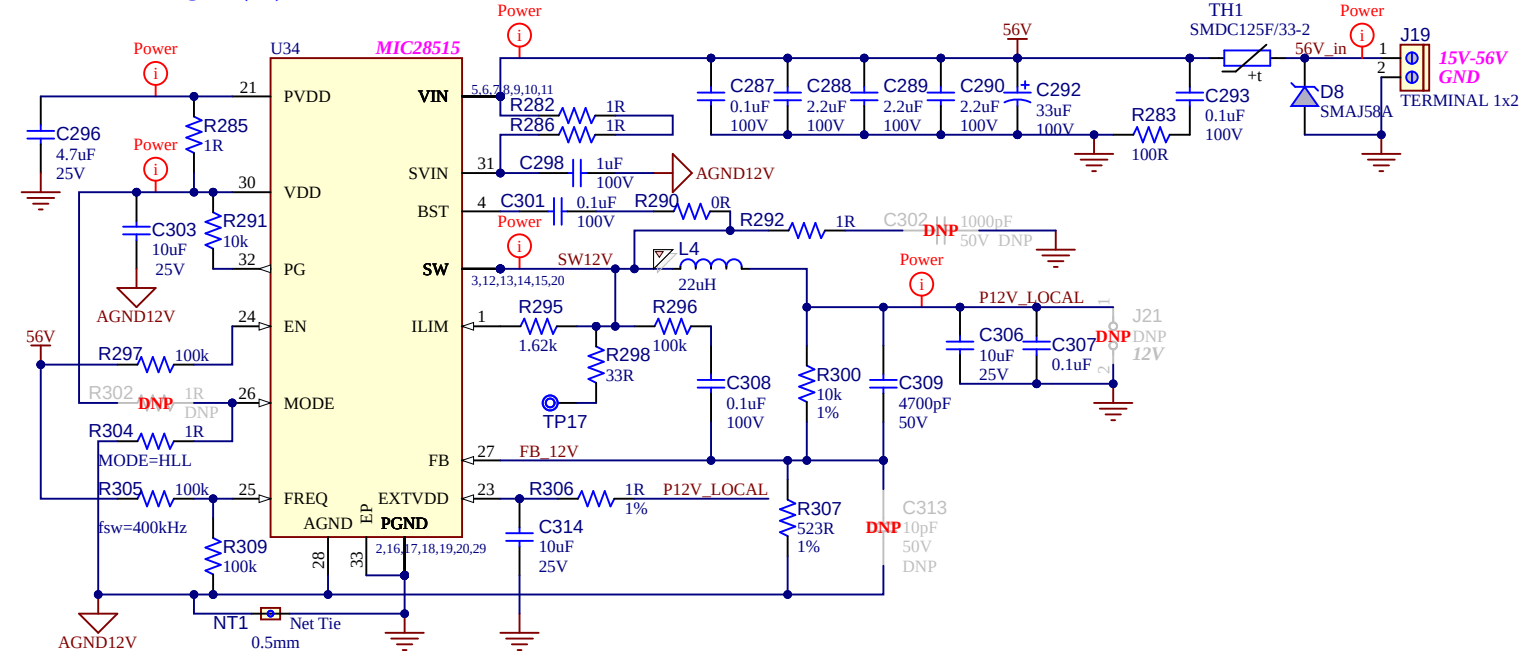


Project Owner: Jorgen Abrahamsen			
PCB Layout Contact: Jorgen Abrahamsen			
PartNumber: EV23X71A	Project Title LAN969x 24-port EVB	Variant: Standard	
Sheet Title LAN9696 Power			
Size A3	SCH #: 02-00733	Rev: 2	Date: 5/17/2024
	PCB #: 04-11815	Rev: 2	Sheet 11 of 13
File: LAN969x 24-port EVB_11-Power.SchDoc			

Designed with

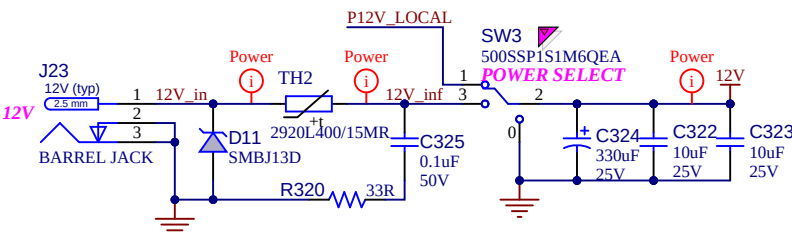
Altium.com

15V-56V in, 12V @ 4A (5A)

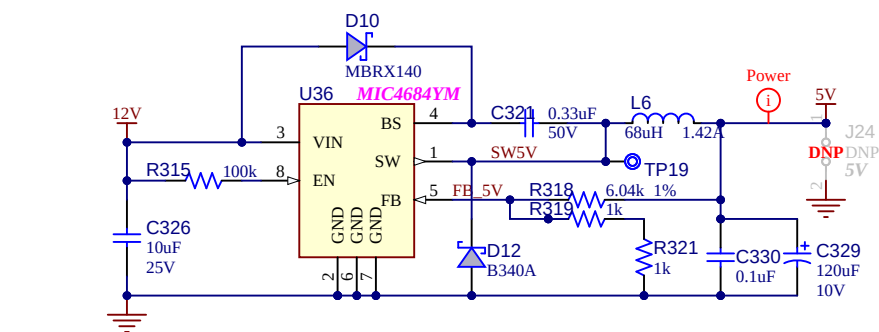


Power input selection

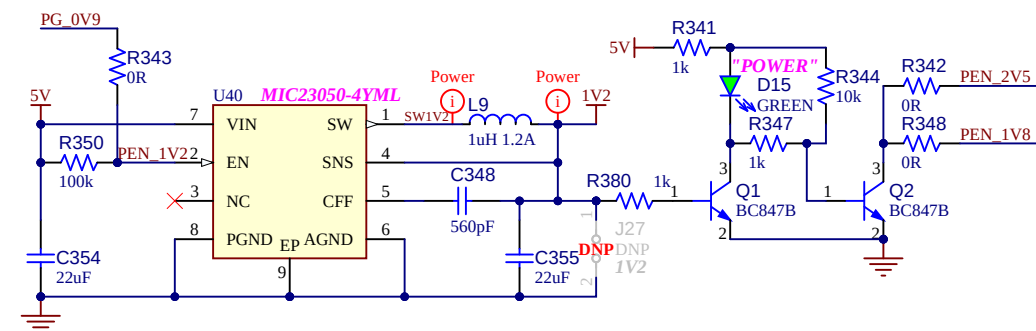
SW3 selects J19 or J23 as power source



5V @ 1.2A (2A)



1V2 @ 480mA (600mA)



1V1 @ 5A (8.5A)

U_UNG8398_C_13-Regulator_Modules
LAN969x 24-port EVB_13-Regulator_Modules.SchDoc



3V3 @ 5A (8.5A)

U_UNG8398_C_13-Regulator_Modules
LAN969x 24-port EVB_13-Regulator_Modules.SchDoc



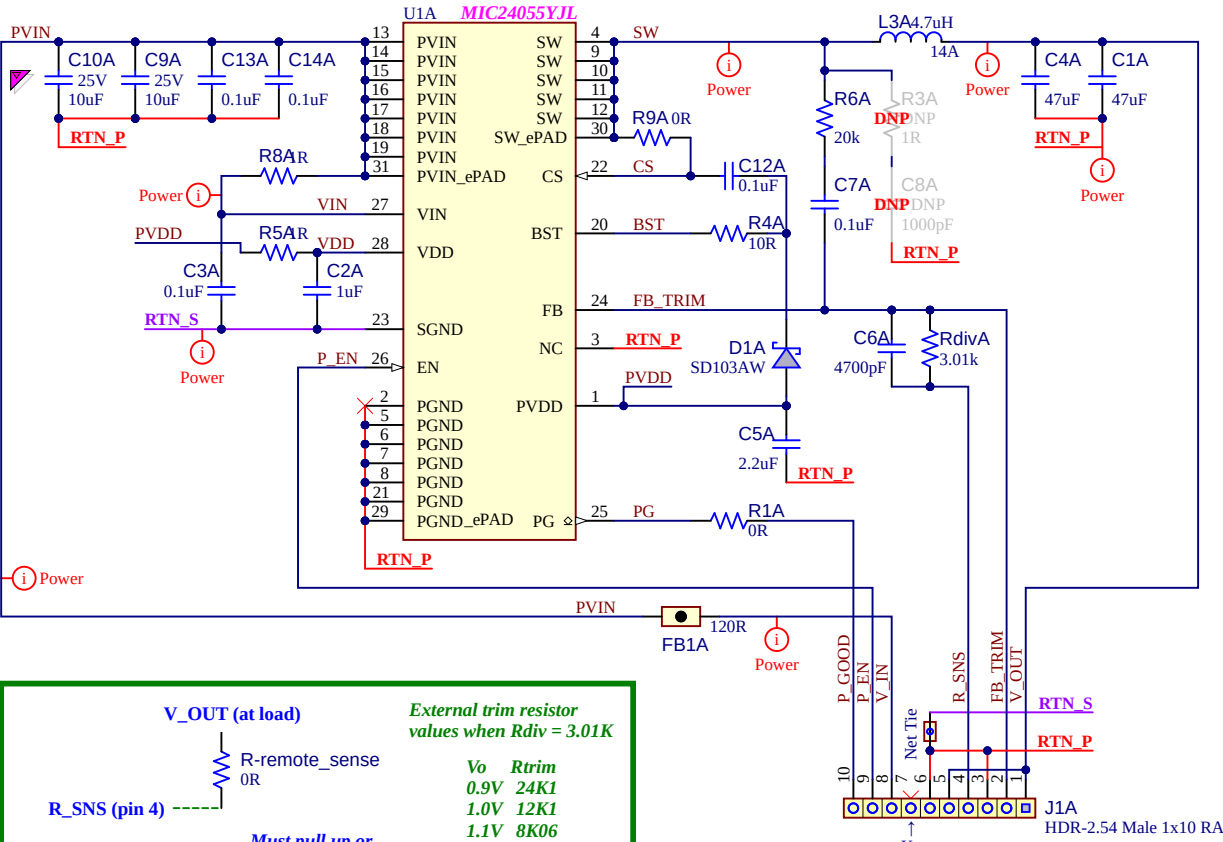
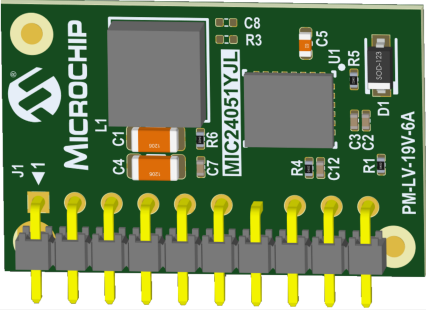
0V9 @ 7A (8.5A)

U_UNG8398_C_13-Regulator_Modules
LAN969x 24-port EVB_13-Regulator_Modules.SchDoc



PM-LV-19V-6A (MIC24055 Power Module)

- Pin list:
- 1. Vo. Output voltage. 0.8V to 5V. 8.5A maximum
 - 2. FB_Trim. Voltage trim / Feedback node.
Place a resistor on the host board between this pin and pin 6 to set Vo.
 - 3. RTN_P = Power Return
 - 4. SNS. Remote sense, Vo sample for FB resistor network.
Use a 0Ω tap from Vo on the target board, or short pins 4 and 5.
 - 5. Vo. (same as pin 1)
 - 6. RTN_P = Power Return & tie point to isolated RTN_S
 - 7. Key
 - 8. Vi. Input voltage. 4.5V to 19V
 - 9. POWER_ENABLE. Active high input. Vih = 1.8V to Vi. Vil = 0.6V.
Must pull up or drive high.
 - 10. POWER_GOOD. Open drain, active high output.
Put a 100kΩ pull-up on the host board if used.



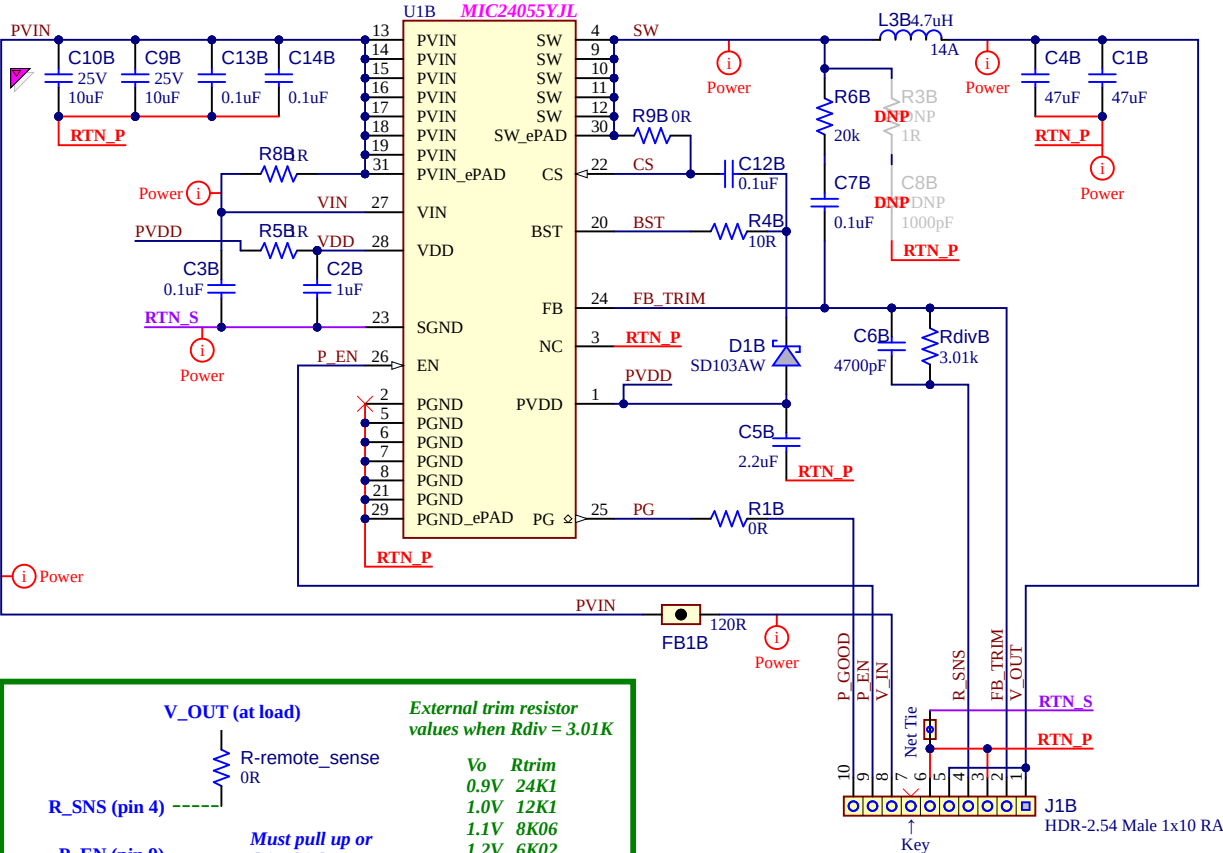
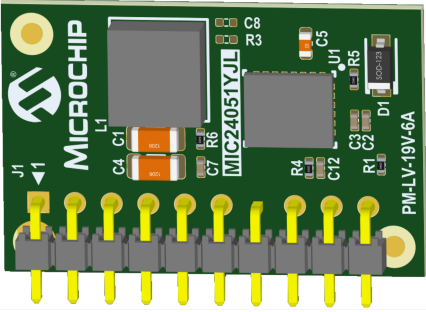
V_OUT (at load)		External trim resistor values when Rdiv = 3.01K	
R-remote_sense	0R	Vo	Rtrim
R_SNS (pin 4)	-----	0.9V	24K1
P_EN (pin 9)	-----	1.0V	12K1
	-----	1.1V	8K06
	-----	1.2V	6K02
	-----	1.35V	4K38
FB_TRIM (pin 2)	-----	1.5V	3K44
	-----	1.8V	2K41
	-----	2.5V	1K42
RTN_P (pin 6)	-----	3.3V	963R

For best results, connect R-trim to pin 6, rather than to pin 3.



PM-LV-19V-6A (MIC24055 Power Module)

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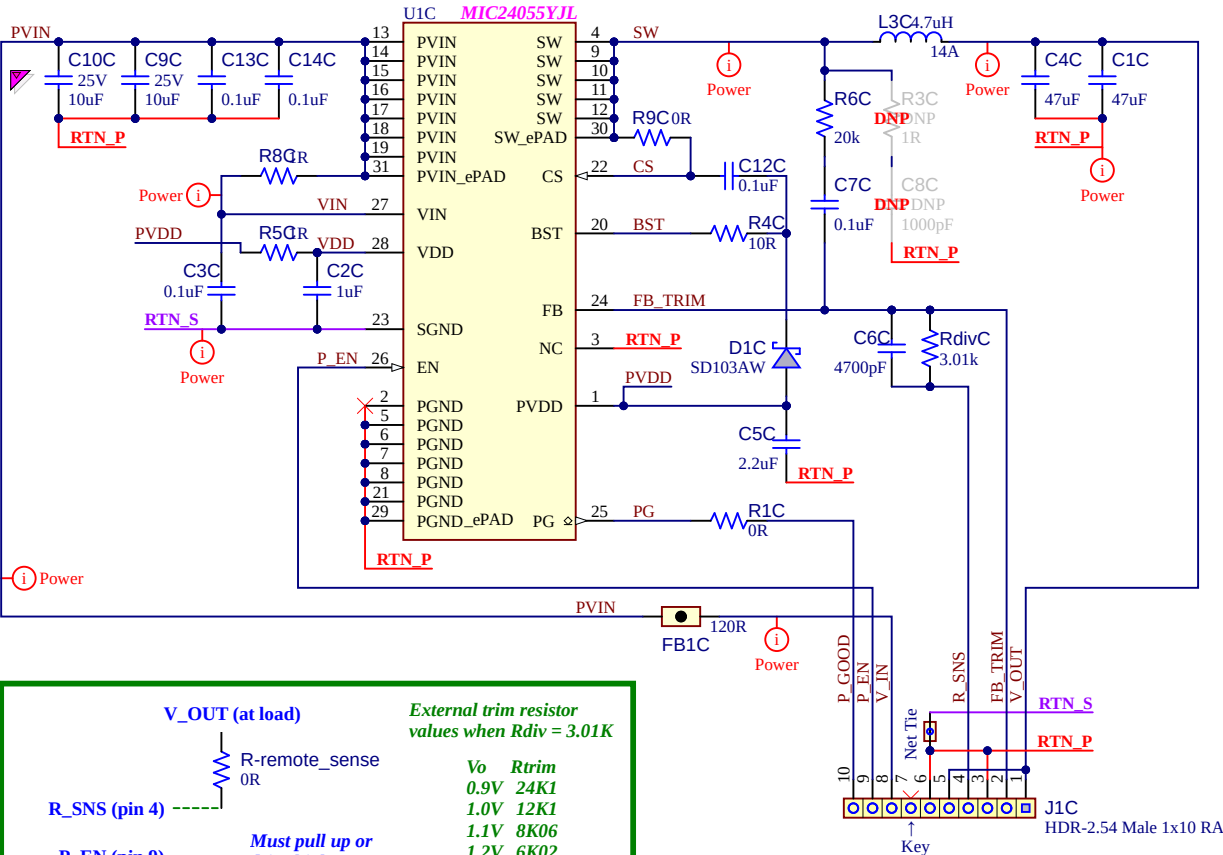
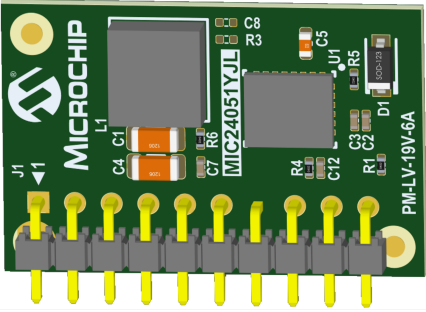
Project Owner: Jorgen Abrahamsen			
PCB Layout Contact: Jorgen Abrahamsen			
PartNumber: EV23X71A	Project Title LAN969x 24-port EVB	Variant: Standard	
Sheet Title Regulator Modules			
Size A3	SCH #: 02-00733	Rev: 2	Date: 5/17/2024
	PCB #: 04-11815	Rev: 2	Sheet 13 of 13
File: LAN969x 24-port EVB_13-Regulator_Modules.SchDoc			

Designed with

Altium.com

PM-LV-19V-6A (MIC24055 Power Module)

- Pin list:
- 1. Vo. Output voltage. 0.8V to 5V. 8.5A maximum
 - 2. FB_Trim. Voltage trim / Feedback node.
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